

The Complete Treatise on Distributional Fixed Points and Cumulant-Field Dynamics of OHLC Processes

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Abstract

This treatise develops a compact mathematical framework for modelling open, high, low, and close (OHLC) processes as self-consistent distributional dynamical systems. Instead of treating the four prices as unrelated scalar coordinates, the framework decomposes an OHLC observation into range and close-location variables, constructs moments and joint cumulants from a rolling stochastic environment, and feeds those distributional objects back into a nonlinear state-transition operator. The resulting system is naturally expressed as a distributional fixed-point problem.

The framework connects fixed-point theory, stochastic processes, time-series econometrics, financial economics, statistical mechanics, information theory, control theory, machine learning, behavioural modelling, biological homeostasis, and complex adaptive systems. Particular attention is given to OHLC admissibility, cumulant hierarchies, local stability, identification, overfitting, empirical implementation, and computational algorithms.

The objective is not to assert that a particular cumulant specification is empirically correct. Rather, the objective is to establish a mathematically coherent class of models that can subsequently be estimated, falsified, and compared with conventional financial time-series models.

The treatise ends with “The End”

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1 Introduction

Financial prices are commonly represented through the quadruple

$$X_t = \begin{pmatrix} O_t \\ H_t \\ L_t \\ C_t \end{pmatrix}. \quad (1)$$

A conventional time-series model may regard these observations as a vector of variables whose conditional dynamics depend on lags, covariates, latent factors, or stochastic volatility. The present framework asks a different question:

Can the next OHLC state be generated self-consistently from the distributional properties of the recent OHLC process itself?

The question leads naturally to a fixed-point equation

$$X_* = F(X_*, K_*), \quad (2)$$

where K_* is a vector of moments or cumulants generated by the distribution associated with X_* .

This formulation has connections with fixed-point theory [4, 8], stochastic-process theory [6], time-series econometrics [13], financial econometrics [11, 7], and statistical mechanics [19].

The central claim of the treatise is therefore structural rather than empirical: OHLC dynamics can be represented as a coupled system in which market geometry generates distributional information and distributional information feeds back into market geometry.

2 OHLC Geometry

Define the intraperiod range

$$R_t = H_t - L_t \quad (3)$$

and the open-to-close displacement

$$B_t = C_t - O_t. \quad (4)$$

Define also

$$U_t = H_t - O_t, \quad (5)$$

$$D_t = O_t - L_t. \quad (6)$$

Then

$$R_t = U_t + D_t, \quad (7)$$

$$B_t = U_t - D_t. \quad (8)$$

Consequently,

$$U_t = \frac{R_t + B_t}{2}, \quad (9)$$

$$D_t = \frac{R_t - B_t}{2}. \quad (10)$$

For $R_t > 0$, define the normalized close-location variable

$$q_t = \frac{B_t}{R_t}. \quad (11)$$

For an admissible OHLC observation,

$$-1 \leq q_t \leq 1. \quad (12)$$

The entire candle can therefore be reconstructed from (O_t, R_t, q_t) :

$$C_t = O_t + R_t q_t, \quad (13)$$

$$H_t = O_t + \frac{R_t}{2}(1 + q_t), \quad (14)$$

$$L_t = O_t - \frac{R_t}{2}(1 - q_t). \quad (15)$$

This transformation separates two economically distinct dimensions:

1. R_t measures scale or intraperiod dispersion;
2. q_t measures directional placement of the close within the observed range.

The decomposition is closely related to range-based volatility measurement [18, 20, 25].

3 The Distributional State

Consider a rolling window

$$\mathcal{W}_t = \{(R_{t-j}, q_{t-j}) : j = 0, \dots, n-1\}. \quad (16)$$

Let the corresponding random vector be

$$Y_t = \begin{pmatrix} R_t \\ q_t \end{pmatrix}. \quad (17)$$

The distributional state can be summarized through moments or, preferably for many purposes, cumulants.

Let

$$K_t(\theta) = \log E_t [\exp (\theta^\top Y_t)] \quad (18)$$

denote the cumulant-generating function, whenever it exists in a neighbourhood of the origin.

The multivariate cumulants are obtained through

$$\kappa_{ab,t} = \left. \frac{\partial^{a+b} K_t(\theta)}{\partial \theta_1^a \partial \theta_2^b} \right|_{\theta=0}. \quad (19)$$

A finite fourth-order representation contains

$$\kappa_{10}, \quad \kappa_{01}, \quad (20)$$

$$\kappa_{20}, \quad \kappa_{11}, \quad \kappa_{02}, \quad (21)$$

$$\kappa_{30}, \quad \kappa_{21}, \quad \kappa_{12}, \quad \kappa_{03}, \quad (22)$$

$$\kappa_{40}, \quad \kappa_{31}, \quad \kappa_{22}, \quad \kappa_{13}, \quad \kappa_{04}. \quad (23)$$

The second-order cumulants encode variance and covariance, third-order cumulants encode asymmetry, and fourth-order cumulants encode higher-order tail and dependence information. These interpretations follow the standard theory of cumulants and multivariate distributions [16, 17].

4 Distributional Fixed Points

Definition 1. *A distributional OHLC fixed point is a state X_* satisfying*

$$X_* = F(X_*, K_*), \quad (24)$$

where K_* is the vector of moments or cumulants generated by the distribution associated with the stationary state X_* .

Written componentwise,

$$O_* = f(X_*, K_*), \quad (25)$$

$$H_* = g(X_*, K_*), \quad (26)$$

$$L_* = h(X_*, K_*), \quad (27)$$

$$C_* = i(X_*, K_*). \quad (28)$$

The essential distinction from an ordinary algebraic fixed point is that K_* is not an arbitrary parameter. It is generated by the state itself.

Thus the system has the self-consistency structure

$$X_* = F(X_*, K(X_*)). \quad (29)$$

This is analogous in structure to self-consistency equations in statistical physics, equilibrium models, mean-field systems, and dynamic programming [19, 5].

5 A Cumulant-Driven OHLC Model

Define

$$s_t = \sqrt{\kappa_{20,t}}. \quad (30)$$

Define standardized skewness and excess kurtosis by

$$\gamma_{1,t} = \frac{\kappa_{30,t}}{\kappa_{20,t}^{3/2}}, \quad (31)$$

$$\gamma_{2,t} = \frac{\kappa_{40,t}}{\kappa_{20,t}^2} - 3. \quad (32)$$

Define the correlation-type quantity

$$\rho_t = \frac{\kappa_{11,t}}{\sqrt{\kappa_{20,t} \kappa_{02,t}}}. \quad (33)$$

A compact nonlinear specification is

$$O_{t+1} = O_t + a_1 s_t + a_2 s_t \gamma_{1,t} + a_3 s_t \gamma_{2,t} + a_4 s_t \rho_t, \quad (34)$$

$$C_{t+1} = C_t + b_1 s_t + b_2 s_t \gamma_{1,t} + b_3 s_t \gamma_{2,t} + b_4 s_t \rho_t, \quad (35)$$

$$R_{t+1} = R_t + c_1 s_t + c_2 s_t |\gamma_{1,t}| + c_3 s_t \gamma_{2,t} + c_4 R_t \rho_t^2, \quad (36)$$

and

$$q_{t+1} = \tanh(d_1 q_t + d_2 \gamma_{1,t} + d_3 \rho_t + d_4 \gamma_{2,t} q_t). \quad (37)$$

The OHLC variables are reconstructed through

$$O_{t+1} = O_{t+1}, \quad (38)$$

$$H_{t+1} = O_{t+1} + \frac{R_{t+1}}{2}(1 + q_{t+1}), \quad (39)$$

$$L_{t+1} = O_{t+1} - \frac{R_{t+1}}{2}(1 - q_{t+1}), \quad (40)$$

$$C_{t+1} = O_{t+1} + R_{t+1} q_{t+1}. \quad (41)$$

The identity notation in the first equation emphasizes that the open is the primitive location coordinate in this particular parameterization.

6 OHLC Admissibility

Proposition 1. *Suppose*

$$R_t > 0 \quad (42)$$

and

$$q_t \in [-1, 1]. \quad (43)$$

Then the reconstructed OHLC vector satisfies

$$H_t \geq O_t, \quad (44)$$

$$L_t \leq O_t, \quad (45)$$

$$H_t \geq C_t, \quad (46)$$

$$L_t \leq C_t. \quad (47)$$

Proof. From

$$H_t - O_t = \frac{R_t}{2}(1 + q_t), \quad (48)$$

the right-hand side is nonnegative because $R_t > 0$ and $q_t \geq -1$. Similarly,

$$O_t - L_t = \frac{R_t}{2}(1 - q_t) \geq 0. \quad (49)$$

Finally,

$$C_t - L_t = \frac{R_t}{2}(1 + q_t) \geq 0, \quad (50)$$

$$H_t - C_t = \frac{R_t}{2}(1 - q_t) \geq 0. \quad (51)$$

Hence the OHLC inequalities follow. $\square$

The use of $\tanh$ in the proposed model is therefore not merely a machine learning convenience. It provides a smooth map from $\mathbb{R}$ into $(-1, 1)$ and consequently enforces the close-location constraint.

7 Seven Distributional Model Classes

The general framework admits many distinct specifications.

Table 1: Representative cumulant-driven model classes

Model	Distributional inputs	Primary interpretation
I	κ_2	Scale and covariance
II	κ_2, κ_3	Scale and asymmetry
III	κ_2, κ_4	Scale and tail structure
IV	$\kappa_2, \kappa_3, \kappa_{21}, \kappa_{12}$	Cross-asymmetry
V	$\kappa_2, \kappa_3, \kappa_4, \kappa_{22}$	Higher-order interaction
VI	$K_t, K_{t+1} = G(K_t)$	Recursive distributional dynamics
VII	$\{\kappa_\alpha\}_{ \alpha \geq 1}$	Full cumulant hierarchy

These should be regarded as nested hypotheses rather than as seven claims that one specification is universally superior to another.

8 Fixed-Point Conditions

At a stationary point,

$$R_{t+1} = R_t = R_*, \quad (52)$$

$$q_{t+1} = q_t = q_*. \quad (53)$$

The range equation therefore implies

$$c_1 s_* + c_2 s_* |\gamma_{1,*}| + c_3 s_* \gamma_{2,*} + c_4 R_* \rho_*^2 = 0. \quad (54)$$

The close-location equation implies

$$q_* = \tanh(d_1 q_* + d_2 \gamma_{1,*} + d_3 \rho_* + d_4 \gamma_{2,*} q_*). \quad (55)$$

The equilibrium is consequently determined by simultaneous restrictions on scale, asymmetry, dependence, tail behaviour, range, and close location.

9 Local Stability

Let

$$Y_t = \begin{pmatrix} R_t \\ q_t \\ K_t \end{pmatrix} \quad (56)$$

and suppose

$$Y_{t+1} = \Phi(Y_t). \quad (57)$$

At the fixed point,

$$\Phi(Y_*) = Y_*. \quad (58)$$

A first-order expansion gives

$$Y_{t+1} - Y_* \approx D\Phi(Y_*)(Y_t - Y_*). \quad (59)$$

Hence a standard sufficient local condition for asymptotic stability is

$$\rho(D\Phi(Y_*)) < 1, \quad (60)$$

where $\rho(\cdot)$ denotes spectral radius.

This places the model within the standard theory of nonlinear dynamical systems and local stability [22, 14].

10 Econometric Interpretation

The framework can be interpreted as a nonlinear conditional distribution model. Let $\mathcal{F}_t$ denote the information set at time t . Then

$$P(X_{t+1} \mid \mathcal{F}_t) = P(X_{t+1} \mid X_t, K_t). \quad (61)$$

The cumulants become conditional distributional statistics:

$$\kappa_{r,t} = \kappa_r(X_{t+1} \mid \mathcal{F}_t). \quad (62)$$

This creates a natural connection to ARCH and GARCH models [11, 7], stochastic volatility [23], realized volatility, and high-frequency econometrics [1].

The proposed framework differs from standard volatility models because volatility is not necessarily the sole state variable. Skewness, kurtosis, cross-cumulants, range, and close-location information can simultaneously enter the transition law.

11 Financial Interpretation

The variables admit several financial interpretations.

Table 2: Financial interpretation of the principal state variables

Variable	Statistical interpretation	Financial interpretation
R_t	Range	Intraperiod dispersion
q_t	Normalized location	Directional close strength
κ_2	Variance	Risk/volatility
κ_3	Third cumulant	Asymmetry/skewness
κ_4	Fourth cumulant	Tail/peakedness information
κ_{11}	Covariance	Joint dependence
ρ	Correlation	Co-movement

The model therefore separates at least three concepts often conflated in simple financial models:

1. scale of uncertainty;
2. asymmetry of uncertainty;
3. dependence among dimensions of uncertainty.

This is relevant to asset pricing, risk management, derivatives, portfolio construction, and market microstructure.

12 Information-Theoretic Interpretation

The cumulant-generating function

$$K(\theta) = \log E[e^{\theta^\top Y}] \quad (63)$$

is closely related to exponential-family representations and large-deviation theory [9, 10].

The derivatives

$$\nabla K(0), \quad \nabla^2 K(0), \quad \nabla^3 K(0), \dots \quad (64)$$

successively reveal increasingly detailed properties of the distribution.

Thus the proposed system may be interpreted as a compressed information architecture:

$$\text{Recent OHLC data} \longrightarrow \text{distributional information} \longrightarrow \text{state transition}. \quad (65)$$

The danger is equally clear: higher-order cumulants can contain substantial estimation noise. A finite-order truncation should therefore be treated as a model-selection problem rather than as an automatic improvement.

13 Statistical Mechanics Analogy

There is a structural analogy with statistical mechanics.

A microscopic state generates macroscopic observables; macroscopic observables then determine effective equilibrium conditions. Here, individual OHLC observations generate empirical distributional quantities, while those distributional quantities determine the effective next-state dynamics.

Schematically,

$$\text{microstates} \rightarrow \text{distribution} \rightarrow \text{macroscopic cumulants} \rightarrow \text{effective dynamics.} \quad (66)$$

The analogy should not be interpreted as claiming that financial markets are physical thermodynamic systems. It is a mathematical analogy concerning aggregation, self-consistency, and equilibrium.

14 Control-Theoretic Interpretation

Write the model as

$$Y_{t+1} = \Phi(Y_t, u_t), \quad (67)$$

where u_t represents exogenous information.

The cumulant vector can then be regarded as a state-dependent sufficient feature vector for a control system:

$$u_t = \Psi(K_t, Z_t), \quad (68)$$

where Z_t represents macroeconomic, order-flow, fundamental, or policy variables.

This opens a path toward robust control, adaptive control, and model predictive control [3].

15 Behavioural and Biological Analogies

The framework also admits disciplined interdisciplinary analogies.

In behavioural economics, agents may react not only to expected outcomes but also to perceived dispersion, asymmetry, and tail risk. Higher cumulants can therefore represent distributional features of perceived uncertainty.

In biology, homeostatic systems maintain variables through feedback:

$$\text{state} \rightarrow \text{sensed deviation} \rightarrow \text{feedback} \rightarrow \text{corrected state.} \quad (69)$$

The OHLC cumulant system has the analogous abstract structure

$$\text{market state} \rightarrow \text{distributional measurement} \rightarrow \text{feedback} \rightarrow \text{new market state.} \quad (70)$$

This does not imply that financial markets possess biological agency. The analogy concerns feedback architecture.

16 Psychological and Psychiatric Analogy

A further analogy can be made with models of perception and uncertainty: agents may respond differently to variance, skewness, and rare-event probability. Prospect theory and related behavioural models demonstrate that distributions cannot always be reduced to their means [15].

A scientifically responsible model should nevertheless distinguish between observable market statistics and latent psychological constructs. The cumulants are directly statistical; psychological interpretations require separate empirical identification.

17 Philosophical Interpretation

The model illustrates a general epistemological distinction between an object and its description.

The OHLC path is an observable representation. The cumulant field is a compressed representation of the distribution generated by observations. The transition operator maps that representation into a new state.

Thus,

$$X \rightarrow \mathcal{D}(X) \rightarrow K(\mathcal{D}(X)) \rightarrow F(K(\mathcal{D}(X))) \quad (71)$$

is a hierarchy of representations.

The fixed point asks whether the representation reproduces the object:

$$X_* = F(K(\mathcal{D}(X_))). \quad (72)$$

This is closely related to the general mathematical idea of self-reference through an operator rather than through a linguistic paradox.

18 Political Economy and Geopolitical Extensions

The framework can be extended from financial markets to macro-financial systems.

Let

$$Z_t = \begin{pmatrix} X_t \\ M_t \\ F_t \\ G_t \end{pmatrix}, \quad (73)$$

where M_t represents monetary variables, F_t fiscal variables, and G_t geopolitical or external variables.

Then

$$Z_{t+1} = \mathcal{F}(Z_t, K_t, \Xi_t), \quad (74)$$

where Ξ_t represents exogenous shocks.

The same architecture can represent interactions among financial risk, monetary policy, fiscal policy, trade conditions, sanctions, conflict risk, and international capital flows.

Such an extension should not be interpreted as establishing causal relationships merely because the mathematical representation permits them. Identification remains an empirical question.

19 Welfare Economics

Suppose an equilibrium distribution determines consumption, wealth, or employment outcomes. Let welfare be

$$W_t = \mathcal{W}(E_t[C_t], \text{Var}_t(C_t), \text{Skew}_t(C_t), \text{Kurt}_t(C_t)). \quad (75)$$

Then two states with identical expected consumption can produce different welfare when their risk distributions differ.

This provides a natural connection between distributional macroeconomics, risk-sensitive welfare analysis, and inequality measurement.

20 Warfare Economics and Strategic Risk

The same mathematical structure can represent high-dimensional strategic risk without assuming that the financial market itself is the strategic actor.

Let

$$S_t = \begin{pmatrix} E_t \\ R_t \\ C_t \\ D_t \end{pmatrix} \quad (76)$$

represent economic exposure, resource capacity, conflict-related risk, and defensive capacity.

A cumulant-field representation can then characterize uncertainty in strategic outcomes:

$$S_{t+1} = F(S_t, K_t, \Xi_t). \quad (77)$$

The appropriate interpretation is that distributional methods can quantify uncertainty surrounding strategic states. They do not themselves determine political objectives or military decisions.

21 Machine Learning Interpretation

Let the cumulant vector be a feature map

$$\phi_t = \Phi(K_t). \quad (78)$$

A neural architecture can then approximate

$$X_{t+1} = F_\theta(X_t, \phi_t). \quad (79)$$

A dual architecture may separate temporal memory and distributional representation:

$$h_t^{(S)} = \text{SSM}(X_{1:t}), \quad (80)$$

$$h_t^{(T)} = \text{Transformer}(X_{1:t}), \quad (81)$$

$$z_t = \text{Concat}\left(h_t^{(S)}, h_t^{(T)}, \phi_t\right), \quad (82)$$

$$X_{t+1} = F_\theta(z_t). \quad (83)$$

This creates a natural bridge between state-space models, attention architectures, feature engineering, and distributional modelling [24, 12].

The cumulant layer should not be assumed to outperform learned representations without out-of-sample evidence.

22 Overfitting and Identification

The flexibility of cumulant systems creates a major statistical danger.

The number of multivariate cumulants grows rapidly with order. If the parameter vector is

$$\theta \in \mathbb{R}^p, \quad (84)$$

then increasing cumulant order can make p large relative to the effective sample size.

Consequently, one should compare nested models using out-of-sample evaluation, penalized likelihood, cross-validation, information criteria, or Bayesian model comparison [2, 21].

A useful hierarchy is

$$\mathcal{M}_2 \subset \mathcal{M}_3 \subset \mathcal{M}_4 \subset \cdots. \quad (85)$$

The empirical question is not whether higher-order cumulants can improve in-sample fit. They almost certainly can. The question is whether they provide stable predictive information outside the estimation sample.

23 Empirical Identification Strategy

A practical empirical study should proceed in stages.

1. Construct synchronized OHLC observations.
2. Remove or document bad ticks, splits, corporate actions, and missing observations.
3. Compute R_t , B_t , and q_t .
4. Estimate rolling moments and cumulants.
5. Estimate low-order models first.
6. Introduce higher-order cumulants sequentially.
7. Compare predictive performance on untouched data.
8. Test parameter stability across regimes.
9. Examine residual dependence.
10. Perform robustness tests across assets, frequencies, and market regimes.

A suitable baseline should include conventional models such as ARIMA, GARCH-type specifications, stochastic volatility models, and appropriate range-based volatility models.

24 Algorithmic Procedure

The following pseudocode describes the basic estimation pipeline.

INPUT:

OHLC observations
rolling window n
maximum cumulant order K
model class M

FOR each time t after the initial window:

1. Read OHLC observations in the rolling window.

2. Compute:

$$R = H - L$$

$$B = C - O$$

$$q = B / R$$

3. Validate:

$$R > 0$$

$$-1 \leq q \leq 1$$

4. Estimate moments/cumulants:

$$K_t = \text{CUMULANTS}(R, q, \text{order} = K)$$

5. Construct feature vector:

$$\Phi_t = \text{NORMALIZE}(K_t)$$

6. Predict:

$$\hat{X}_{t+1} = F_{\theta}(X_t, \Phi_t)$$

7. Reconstruct OHLC:

$$\hat{H}_t = \hat{O}_t + \hat{R}_t(1 + \hat{q}_t)/2$$

$$\hat{L}_t = \hat{O}_t - \hat{R}_t(1 - \hat{q}_t)/2$$

$$\hat{C}_t = \hat{O}_t + \hat{R}_t \hat{q}_t$$

8. Store prediction and residual.

END FOR

Evaluate:

predictive loss
calibration
stability
parameter robustness
regime robustness
comparison with benchmark models

25 Architecture Diagram

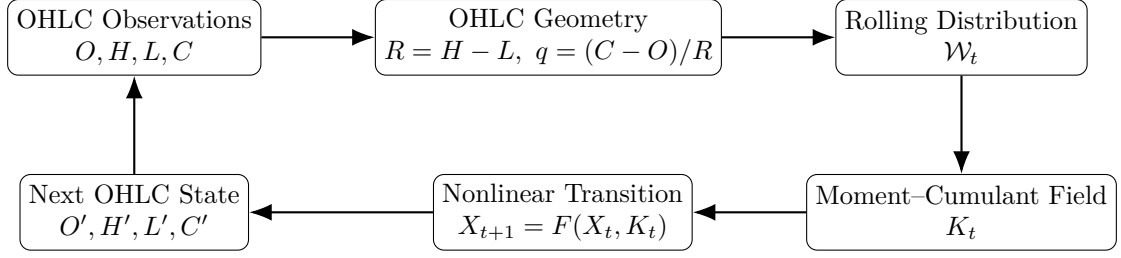


Figure 1: Distributional fixed-point architecture for OHLC dynamics.

26 Physical, Chemical, and Biological Analogies

The mathematical architecture appears in several scientific disciplines.

In physics, macroscopic equilibrium can emerge from microscopic states through partition functions and cumulant-generating structures [19].

In chemistry, equilibrium systems are governed by coupled nonlinear constraints involving concentrations, reaction rates, and thermodynamic potentials.

In biology, feedback and homeostasis generate self-correcting dynamical systems.

In each case the relevant abstract pattern is

$$\text{state} \rightarrow \text{summary statistics} \rightarrow \text{feedback} \rightarrow \text{new state.} \quad (86)$$

The OHLC framework adopts this mathematical pattern without asserting that financial markets obey physical or biological laws.

27 Model Risk

Several failure modes require explicit attention.

Table 3: Principal model risks

Risk	Description
Sampling error	Higher cumulants are noisy, especially in short windows.
Nonstationarity	Market distributions can change over time.
Microstructure noise	Bid-ask effects and discrete prices contaminate high-frequency statistics.
Look-ahead bias	Rolling statistics must use only information available at prediction time.
Overfitting	Flexible cumulant expansions can fit noise.
Regime dependence	Parameters estimated in one regime may fail in another.
Numerical instability	High-order cumulants can be poorly conditioned.
Identification failure	Predictive association does not establish economic causality.

These risks are not peripheral. They determine whether the mathematical construction has empirical value.

28 A General Cumulant-Field Formulation

Let

$$\mathcal{K}_t = \{\kappa_{\alpha,t} : |\alpha| \geq 1\} \quad (87)$$

denote the cumulant field.

The general model is

$$X_{t+1} = \mathcal{F}(X_t, \mathcal{K}_t, Z_t, \Xi_t), \quad (88)$$

where Z_t represents observable exogenous variables and Ξ_t represents innovations.

The distributional state satisfies

$$\mathcal{K}_t = \mathcal{K}(\mathcal{D}_t), \quad (89)$$

where $\mathcal{D}_t$ is the conditional distribution of the underlying process.

At equilibrium,

$$X_* = \mathcal{F}(X_*, \mathcal{K}(\mathcal{D}_*), Z_*, \Xi_*). \quad (90)$$

This is the central fixed-point equation of the treatise.

29 Theoretical Proposition

Proposition 2. *Suppose F is continuously differentiable in a neighbourhood of X_* and the cumulant mapping $K(X)$ is continuously differentiable there. Define*

$$T(X) = F(X, K(X)). \quad (91)$$

If

$$T(X_*) = X_* \quad (92)$$

and

$$\rho(DT(X_*)) < 1, \quad (93)$$

then X_* is a locally asymptotically stable fixed point of the induced distributional dynamical system.

Proof. By the chain rule,

$$DT(X) = F_X(X, K(X)) + F_K(X, K(X))DK(X). \quad (94)$$

A first-order Taylor expansion around X_* gives

$$T(X) - X_* = DT(X_*)(X - X_*) + o(\|X - X_*\|). \quad (95)$$

The spectral-radius condition implies local contraction under an equivalent norm, yielding local asymptotic stability by the standard theory of discrete dynamical systems [22]. $\square$

30 Research Programme

The framework suggests a sequence of increasingly demanding empirical tests.

1. Test whether range and close-location variables contain more stable distributional information than raw OHLC levels.
2. Test second-order cumulants.
3. Add third-order cumulants.
4. Add fourth-order cumulants.
5. Test cross-cumulants.
6. Compare fixed-window and adaptive-window estimators.
7. Test regime-dependent parameters.
8. Compare parametric and machine-learning transition operators.
9. Test cross-asset transfer.
10. Conduct genuine out-of-sample evaluation.

A successful result would require not merely statistical significance but economic significance, robustness, stability, and reproducibility.

31 Conclusion

The central construction can be summarized as

$$\boxed{OHLC \xrightarrow[\text{geometry}]{\text{range/location}} \text{distribution} \rightarrow \text{moments/} \xrightarrow[\text{cumulants}]{\text{nonlinear}} \text{feedback} \rightarrow OHLC.} \quad (96)$$

The resulting model is neither merely a price equation nor merely a volatility model. It is a distributional dynamical system whose state transition depends on statistical properties generated by the state itself.

The fundamental object is therefore

$$\boxed{X_* = F(X_*, K(X_*))}. \quad (97)$$

This equation captures the principal theoretical idea: the equilibrium OHLC state must be simultaneously compatible with the distributional structure that generates the transition law.

The framework is deliberately broad. It can be reduced to conventional time-series models, expanded into high-order cumulant systems, embedded in state-space models, combined with neural architectures, or extended to macro-financial systems. Its empirical validity, however, must always be determined by estimation, identification, out-of-sample testing, and replication rather than by mathematical elegance alone.

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Glossary

Admissible OHLC state

A quadruple satisfying $H \geq \max(O, C)$ and $L \leq \min(O, C)$.

Cumulant

A quantity obtained from derivatives of the logarithm of a moment-generating or cumulant-generating function.

Cumulant-generating function

The function $K(\theta) = \log E[\exp(\theta^\top X)]$.

Cumulant field

The collection of cumulants across orders and, where relevant, dimensions or locations.

Distributional fixed point

A state that reproduces itself under a transition operator whose inputs include distributional statistics generated by that state.

Close-location variable

The normalized quantity $q = (C - O)/(H - L)$.

Excess kurtosis

The standardized fourth cumulant, commonly written as $\kappa_4/\kappa_2^2 - 3$.

Fixed point

A state x_* satisfying $T(x_*) = x_*$.

Moment

An expectation of a power or product of random variables.

OHLC

Open, high, low, and close observations for a financial interval.

Range

The intraperiod difference $H - L$.

Self-consistency

The requirement that the statistical quantities determining a model are themselves generated by the state produced by that model.

Skewness

A standardized third central moment or, equivalently under appropriate conditions, a standardized third cumulant.

Spectral radius

The maximum absolute value of the eigenvalues of a matrix.

Stability

The property that sufficiently small deviations from an equilibrium diminish under the induced dynamics.

The End